Exercise 5A: Bayesian Perspective on Regularisation

As discussed in CASI (Eq. 7.37 and following), regularisation can also be understood from a Bayesian perspective, where the parameter vector θ is viewed as a random variable with a certain prior distribution. Let $f(\mathbf{x}|\theta)$ be the joint probability of all data points in your model. This could, for instance, represent the assumption that the residuals are normally distributed in a linear regression model. In the Bayesian framework, one assumes that the parameter vector θ has a prior distribution $f(\theta)$, and when we look at the data to estimate the parameters, we apply Bayes' rule to obtain the posterior $f(\theta|\mathbf{x})$, that is, the new distribution of the parameters conditioned on the data we've seen:

$$f(\theta|\mathbf{x}) = \frac{f(\mathbf{x}|\theta)\,f(\theta)}{f(\mathbf{x})} \tag{1}$$

We can get a point estimate of the parameters (what is called the *maximum a posteriori* or *MAP* estimate) by taking the maximum of the posterior distribution, or its logarithm:

$$\hat{\theta} = \arg\max_\theta \log f(\theta|\mathbf{x}) \tag{2}$$

a) (*) Show that Eq. 2 has the same form as a maximum-likelihood estimate to which ℓ_2 regularisation has been added if you assume that the prior distribution of the parameters is iid normal with mean 0 and some (common) variance σ^2 (Equation 3). What does the regularisation hyperparameter λ correspond to in this view?

$$f(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{3}$$

Hints: Plug Eq. 1 in Eq. 2 and substitute $f(\theta)$ with the form of the prior given by the specific distributional assumptions we're making. You can leave $f(\theta|\mathbf{x})$ unspecified. Take into consideration that the addition of a constant to the objective function doesn't change where the maxima are (it only changes the values of the maxima, not their location), and that $f(\mathbf{x})$ is constant if the data are given.

b) (*) Show in the same way that ℓ_1 regularisation corresponds to a prior on the parameters given by a zero-mean Laplace distribution.

Hint: The pdf of the Laplace distribution is

$$f(x|\mu,b) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right) \tag{4}$$

c) (*) Create a plot of a Gaussian and a Laplace distribution with zero mean and equal variance (Hint: In terms of the parameter b, the variance of the Laplace distribution is equal to $2b^2$). Compare the shapes of the two distributions. What do the different prior distributions tell us about the parameter values we're likely to obtain from each regularisation method?